Resonances

So far, we've used sines/cosines to describe the motion of strings/vibrating air/waves in general because they seemed like a convenient tool. They are quite a bit more fundamental than that, and I'll go over some of that in this note.

1. Separation of Variables

I'm going to indulge myself in some math that is well beyond the scope of this class. Our end goal is a simple principle that's truly fundamental to how just about any vibrating body works. Don't worry about following the details of the math; just make sure to absorb the end result we'll get.

We're going to go back to the wave equation and look for solutions that look like $Y(x)A(t)$. "We already have perfectly good solutions for the wave equation - $f(x \pm vt)$!" I hear you asking. Why would I want to go from that perfectly simple that depends on the *difference* of position and time to a solution that depends on the *product* of position and time? And can we even express one set of solutions in terms of the other? Well, the $f(x \pm vt)$ doesn't really tell us about just the time behavior, and since our ear hears frequencies we really do want to know about just the time part of the solution. And we can indeed express the two sets of solutions in terms of each other, but how that works will wait until another note.

Solutions of the form $Y(x)A(t)$ are what are called separation of variables solutions (in case you want to read further). They're widely used to solve any partial differential equation (PDE) that is *linear*. That is, if I have two solutions to the PDE, then the sum of those solutions also solves the PDE. That's true of the wave equation, as well as a whole bunch of other equations that show up in physics (including modified versions of the wave equation that better describe real materials).

We're guessing that $f(x,t) = Y(x)A(t)$ solves the wave equation, so we can just plug that in. When we do that we get:

$$\frac{\partial^2 f}{\partial t^2} = c_s^2 \frac{\partial^2 f}{\partial t^2} \tag{1}$$

$$\frac{\partial^2 YA}{\partial t^2} = c_s^2 \frac{\partial^2 YA}{\partial t^2} \tag{2}$$

In words, we're setting the change in slope of our function as we change times to the change in slope as we change position, scaled by the constant c_s^2. You might remember from calculus, or just take my word for it, that as I change time, $A(t)$ changes, but $Y(x)$ doesn't, so I only have to worry about the change in slope of A. Similarly, the change in slope as we change position is only going to affect $Y(x)$ since $A(t)$ doesn't depend on position. We'll call Y'' the change in slope of Y and A'' the change in slope of A. When we do that we get

$$A''Y = c_s^2 Y''A \tag{3}$$

If we divide both sides by our function, we get

$$\frac{A''Y}{AY} = c_s^2 \frac{Y''A}{AY} \tag{4}$$

$$\frac{A''}{A} = c_s^2 \frac{Y''}{Y} \tag{5}$$

Equation **??** has put everything that depends on time on the left, and everything that depends on position on the right. One final bit of gymnastics, and we're nearly there. If Equation **??** holds for all time and for everywhere in space (which it had better) then if I change the time the equation still has to hold. Since the right side didn't change, because it doesn't depend on time, that means the left side can't have changed as I change time. The only way that happens is if the left side is equal to a constant. If the left side is equal to a constant, then the right side had better be equal to the same constant as well.

In general, solving the right side of the equation is *extremely* challenging. At some level, though, we don't care. If we want to know how our instrument vibrates with time, all we need to know is that the right side is a constant. If we take the right side and ignore (for now) the speed of sound, then we can say the constant is $-k^2$ which leaves us with:

$$\frac{Y''}{Y} = -k^2 \tag{6}$$

$$Y'' = -k^2 Y \tag{7}$$

This is the same equation we saw for a Helmholtz resonator. For a simple 1-dimensional string, the solution to this equation is sines/cosines that match the conditions of the string. For instance, if the string is nailed down at both ends, we know that the height of our string has to be zero at the ends. That means our solutions have to look like $\sin(kx)$ but only for values of k that keep the string at zero at the ends. It isn't too bad to solve this in one dimension, but if you go to more dimensions (and I have rarely met a purely 1-dimensional instrument), then Equation **??** gets very much more complicated. The general principle that a given shape can only have certain values of k holds, though. We may have a hard time calcualting what those k's *are*, but we know there's only going to be a limited set of them.

Now we turn our attention to the left side. It equals c_s^2 times $\frac{Y''}{Y}$, but we know that $\frac{Y''}{Y} = -k^2$. If we plug that in, we have:

$$\frac{A''}{A} = -c_s^2 k^2 \tag{8}$$

Define $\omega = c_s k$ and we get

$$\frac{A''}{A} = -\omega^2 \tag{9}$$

or

$$A'' = -\omega^2 A \tag{10}$$

This is once again the equation for the Helmholtz resonator/mass on spring/sipmle harmonic oscillator, and its solutions are $\sin(\omega t)$ and $\cos(\omega t)$.

2. Separation of Variables Discussion

We now have a truly important result. If we have *any* linear PDE where the time part is an acceleration, we'll have solutions where the time part behaves like sine waves. We didn't have to worry *at all* about what the spatial part was doing. If we want to know the frequencies we're going to get, we'll need to solve that tricky right-hand side. We do know that whatever happens, we're going to get some set of overtones. A big part of designing an instrument is shaping it so that you get the overtones you want.

You might also ask "OK, so you found a few special functions where my thing vibrates with a single frequency, but what if I start off withs something different?" For instance, if we pluck a violin string, that does not look like a sine wave. It turns out, though, that we can always write our starting position as the *sum* of our separation of variables solutions. We'll see how to do this for strings in the next note, but we can do it for arbitrary shapes. Since the sum of two solutions is a solution, we can treat our starting shape as the sum of SOV solutions. The details of how we started off tell us how much of each k we have, which in turn tells us how loud the different overtones are. That in turn will set the sound quality of the note. No matter what, though, our object is going to resonate at a fixed set of frequencies. Music is built into the very fabric of nature.